BCAP: Black-Box Cross-Modal Adversarial Patch Attacks Against Multimodal Vehicle Recognition Systems via UAV Remote Sensing

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***Abstract*—The reliability of deep learning-based object detection in uncrewed aerial vehicle (UAV) remote sensing increasingly relies on fusing visible and infrared sensors. However, existing attack methods used to evaluate their robustness are often unrealistic, assuming white-box access or targeting only a single modality. This article first reveals a critical cross-modal vulnerability through region sensitivity analysis on various YOLO detectors, demonstrating that both visible and infrared models exhibit a shared spatial weakness by being most sensitive to perturbations in the target vehicle's central area. Leveraging this key insight, we propose a framework with black-box cross-modal adversarial patch (BCAP) attacks on UAV detectors. BCAP introduces a multiobjective optimization formulation that models visible and infrared modalities as independent yet competing objectives, which enables effective cross-modal optimization without manual weight tuning. It employs a novel grid-constrained Bezier curve generation method to efficiently search for smooth, effective patch shapes within this sensitive central region, further enhanced by an adaptive population restart strategy. Extensive experiments on the DroneVehicle dataset show BCAP achieves a 50%–61% cross-modal attack success rate (ASR), significantly outperforming the 37%–42% range of existing methods. This work provides a more potent and realistic baseline for evaluating the end-to-end reliability of multimodal UAV remote sensing systems.**

***Index Terms*—Adversarial patch attacks, cross-modal, multiobjective optimization, uncrewed aerial vehicle (UAV) remote sensing, vehicle recognition.**

I. Introduction

With the rapid advancement of uncrewed aerial vehicle (UAV) remote sensing technology, deep neural network (DNN)-based UAV object detection has been widely deployed in diverse fields such as wildlife conservation monitoring [1], [2], agricultural crop surveillance [3], and infrastructure inspection [4]. Despite these advancements, the vulnerability of DNNs to adversarial examples has raised serious security concerns [5], [6]. Extensive research has demonstrated that maliciously crafted adversarial perturbations are capable of deceiving various object detection systems [7], [8]. Based on the attacker's knowledge of the target model, adversarial attacks are typically divided into two categories. In white-box attacks, the attacker has complete access to the model's architecture and parameters and can directly perform gradient-based optimization. Conversely, black-box attacks are more aligned with real-world scenarios, assuming that the attacker cannot access any internal information and can only attack by querying the model's outputs. In this context, researchers have begun to explore various stealth attack methods to evaluate the robustness of DNNs in practical applications.

To achieve round-the-clock detection capabilities, modern UAV systems widely adopt visible and infrared sensor fusion technologies [9], [10]. The visible modality provides detailed texture information during daylight, while the infrared modality captures target thermal radiation features at night or in adverse weather conditions. Multimodal systems rely on cross-sensor redundancy mechanisms to enhance overall reliability. Therefore, developing attack methods that can simultaneously deceive both visible and infrared sensors has become key to evaluating the robustness of such fusion systems. A "cross-modal" attack refers to generating a physical perturbation that can simultaneously affect different modalities. Unlike single-modal attacks, such as targeting only the visible modality, cross-modal attacks can directly weaken or even destroy the system's redundancy mechanism, causing the recognition capabilities of all modalities to fail simultaneously. This leads to a complete paralysis of object detection and poses a more severe threat to the secure operation of UAV missions.

However, existing research on adversarial attacks against UAV remote sensing detection systems exhibits significant limitations in scope and applicability. Most UAV-based adversarial attack studies have focused exclusively on single-modal scenarios, targeting either visible or infrared detection systems independently [11], [12], [13]. These approaches fail to address the increasing prevalence of multimodal fusion systems in modern UAV platforms. Conversely, while several cross-modal adversarial attack methods have been developed, they primarily target ground-level detection scenarios such as autonomous driving and pedestrian detection [14], [15], [16]. The unique characteristics of aerial perspectives render these ground-based approaches inadequate for UAV remote sensing detection systems. Furthermore, existing optimization strategies in adversarial attacks typically aggregate multiple

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requirements (such as attack effectiveness and stealth) into a single composite objective function [15], [17] for optimization by algorithms such as differential evolution (DE) [12], [18], [19], rather than treating each modality as an independent optimization target. This research gap leaves UAV-based multimodal detection systems without adequate robustness evaluation frameworks, particularly in black-box scenarios where model parameters remain unknown to attackers.

Based on the above discussion, a cross-modal adversarial attack framework specifically designed for UAV detection systems is developed in this article to address this research gap. In general, the root cause of limited success in UAV cross-modal attacks is the fundamental conflict between optimization objectives of different sensing modalities and the curse of dimensionality in black-box scenarios. The conflicting nature of visible and infrared modality requirements makes traditional single-objective optimization approaches inadequate for simultaneous cross-modal deception. The development of such attacks presents several unique technical challenges that existing methods cannot adequately address. First, the multiscale and multiviewpoint nature of aerial imagery creates significant difficulties in maintaining patch effectiveness across varying altitudes, viewing angles, and target scales [20]. Traditional patches optimized for fixed viewpoints often fail when applied to dynamic UAV remote sensing scenarios. Second, the conflicting optimization objectives between visible and infrared modalities pose fundamental challenges, as these modalities operate on different imaging principles and feature extraction mechanisms [21], which makes simultaneous optimization inherently difficult. Third, black-box attack scenarios compound the complexity by eliminating gradient-based optimization approaches, requiring population-based methods that face curse of dimensionality problems in high-dimensional search spaces [22], [23].

To address these challenges, we propose several innovative technical solutions within our black-box cross-modal adversarial patch (BCAP) framework. For the multiviewpoint robustness challenge, we first conduct comprehensive region sensitivity analysis and discover an important phenomenon: despite visible and infrared modalities operating on different imaging principles, both exhibit similar regional vulnerability distribution patterns in UAV vehicle detection. As illustrated in Fig. 1, central vehicle regions show significantly higher sensitivity to detection perturbations than peripheral areas. Inspired by [18] and [24], we develop a comprehensive region sensitivity analysis that identifies universally vulnerable areas across different scales and viewing angles, focusing patch placement on the central roof region that exhibits 3.2–5.5 times higher sensitivity than peripheral areas. To resolve the cross-modal optimization conflict, we formulate the attack as a multiobjective optimization problem that treats visible and infrared modalities as independent yet conflicting objectives, which allows for simultaneous optimization without manual weight tuning. This formulation is solved using NSGA-II [25], which is validated through systematic comparison with alternative optimization strategies. For the curse of dimensionality in black-box scenarios, we introduce a grid-constrained Bezier curve generation method that significantly reduces the search space while ensuring smooth and effective patch shapes. Additionally, an adaptive restart strategy enhances global search capabilities and prevents premature convergence to local optima.

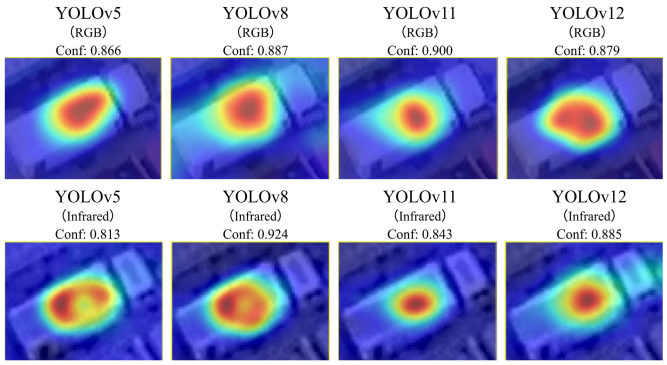


Fig. 1. Central region sensitivity phenomenon in cross-modal UAV vehicle detection.

In summary, the contributions of this article are as follows.

1) To the best of our knowledge, we propose BCAP, the first black-box framework for cross-modal adversarial attacks against UAV remote sensing detection systems, filling a critical gap in the robustness evaluation of aerial multimodal systems.
2) A novel grid-constrained patch generation method is introduced based on region sensitivity analysis, which constructs nine-grid constraints that effectively balance attack effectiveness with computational efficiency.
3) We identify and formulate the cross-modal adversarial attack as a multiobjective optimization problem with inherently conflicting objectives. Systematic comparison of different formulations and optimization strategies validates the necessity of this formulation and demonstrates that it yields significant improvements over single-objective approaches when combined with our grid-constrained patch generation.
4) We conduct comprehensive experimental validation that demonstrates BCAP's effectiveness and robustness across diverse scenarios, with consistent superior performance compared to existing approaches across multiple detector variants and vehicle categories.

II. Related Works

A. Adversarial Patch Attacks

Early adversarial attack methods primarily focused on norm-constrained perturbations, exemplified by FGSM [6], PGD [26], and C&W [27], which add imperceptible noise across entire images to deceive DNNs. To achieve more targeted and controllable attacks, Brown et al. [28] introduced adversarial patches, which concentrate perturbations within localized regions to enable more flexible attack deployment and precise control over perturbation placement. Building upon this foundation, researchers have developed domain-specific patch attack methods for both visible and infrared

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modalities. In the visible domain, Xu et al. [29] design adversarial T-shirts to mislead person detectors, Sharif et al. [30] develop adversarial glasses to evade face recognition systems, and Wei et al. [18] propose using real-world stickers as adversarial patches to deceive face detectors innovatively. Extending adversarial attacks from pedestrians to vehicles, Jiang et al. [31] generate transferable adversarial camouflage on 3-D vehicle surfaces by dispersing multilayer attention through topology-aware perturbation redistribution. Yao et al. [32] further exploit geometric information in 3-D point cloud object tracking by combining point perturbations guided by local curvature with global rotations to generate adversarial samples transferable across different trackers. In the infrared domain, Zhu et al. [8] propose adversarial bulbs and invisible clothing [33] to deceive infrared pedestrian detectors, respectively utilizing additional heat sources and QR code patterns. The two methods are effective. However, they operate under a white-box setting, whereas in real-world applications, attackers cannot access the model's detailed architecture. For this reason, Hu et al. [34] propose a black-box attack method that uses the PSO algorithm to optimize the shape of Bezier curves as patches to attack infrared pedestrian detectors. Recent works have extended these approaches to UAV-based detection scenarios, with Zhang et al. [11] proposing multiscale adversarial patches for aerial imagery. To further enhance attack feasibility and realism, Huang et al. [35] attack UAV remote sensing systems by optimizing multiple small patches within feasible regions to address deployment constraints. Chen et al. [36] introduce a harmonization-based framework to deceive aerial object detectors by optimizing digital-to-physical visual consistency. Jia et al. [37] develop adversarial camouflage coatings to evade remote sensing object detection. However, these approaches generally rely on white-box access to model gradients. In contrast, Tang et al. [12] develop DE-based patch optimization specifically for UAV remote sensing object detection systems in a black-box manner.

Nevertheless, these methods only focus on single modality and cannot work well for cross-modal attacks.

B. Cross-Modal Adversarial Attacks

Cross-modal adversarial attacks involve the introduction of coordinated perturbations across multiple sensing modalities (e.g., visible and infrared), thus deceiving multimodal DNNs into making incorrect detection or classification decisions. Their difficulty partly stems from the heterogeneous visual representations produced by different imaging sources, a broader issue also observed in multiexposure HDR reconstruction, where Hu et al. [38] use semantic knowledge transfer to mitigate SDR/HDR domain and format discrepancies. Wei et al. [39] exploit intermediate features shared by image and video models and adaptively integrate information from multiple layers to generate adversarial videos transferable to black-box video recognition models. In the visible-infrared setting, Kim et al. [40] propose a novel adversarial patch generation framework that exploits Cross-spectral Mapping and Material Emissivity loss to simultaneously attack visible and infrared person detection systems. The following year, Kim et al. [41] design a laminated invisible garment using transparent low-Emissivity films that can simultaneously deceive both visible and infrared person detection systems. Nevertheless, the aforementioned methods assume white-box access to detailed model architectures and parameters, which is typically unavailable in real-world applications. Black-box scenarios, where attackers cannot access internal model information, are more realistic but present greater challenges for cross-modal physical attacks. To address this challenge, Wei et al. [15] design a unified adversarial patch that simultaneously deceives both visible and infrared object detectors under black-box conditions through shape optimization methods. In addition, Liu et al. [42] propose CAMPatch, a two-stage cross-modal adversarial patch attack on visible-infrared pedestrian detectors, which locates salient patch regions with class activation maps in the initialization stage and then refines the patch shape through DE.

The aforementioned cross-modal adversarial attack methods mainly focus on ground-level pedestrian detection scenarios. Long et al. [43] propose CDUPatch, a color-driven universal adversarial patch that exploits color-thermal responses to attack visible-infrared detectors across different scales and viewpoints, extending cross-modal adversarial patch attacks to UAV vehicle scenarios. However, CDUPatch still relies on gradient-based optimization under a white-box setting. Unfortunately, BCAP attacks against UAV-based visible-infrared detection systems remain largely unexplored.

III. Methodology

A. Problem Formulation

In the UAV remote sensing vehicle detection task, given a clean infrared and visible image pair $[I_{\text{vis}}, I_{\text{inf}}]$, the objective of a black-box cross-modal adversarial attack is to generate a patch P, and it creates a set of adversarial samples $[I_{\text{vis}}^{\text{adv}}, I_{\text{inf}}^{\text{adv}}]$ that can make the visible detector $f_{\text{vis}}(\cdot)$ and infrared detector $f_{\text{inf}}(\cdot)$ simultaneously unable to detect the vehicle. The formulation can be expressed as follows:

$$\max\left(f_{\text{vis}}\left(I_{\text{vis}}^{\text{adv}}\right), f_{\text{inf}}\left(I_{\text{inf}}^{\text{adv}}\right)\right) < \varepsilon \tag{1}$$

where $f_{\text{vis}}(I_{\text{vis}}^{\text{adv}})$ and $f_{\text{inf}}(I_{\text{inf}}^{\text{adv}})$ denote the confidence scores of detected vehicle in the visible modality and infrared modality, respectively, and ε is a predefined threshold.

To satisfy the aforementioned attack conditions, we need to specify the generation mechanism of adversarial samples $[I_{\text{vis}}^{\text{adv}}, I_{\text{inf}}^{\text{adv}}]$. Specifically, adversarial samples are created by superimposing the designed adversarial patch P onto designated regions of the original images. This process can be precisely controlled through mask operations, which are mathematically expressed as follows:

$$I_{\text{vis}}^{\text{adv}} = I_{\text{vis}}(1 - M) + P \odot M \tag{2}$$

$$I_{\text{inf}}^{\text{adv}} = I_{\text{inf}}(1 - M) + P \odot M \tag{3}$$

where M denotes a binary mask matrix that defines the placement location and shape of the adversarial patch P in the image, and $\odot$ represents element-wise multiplication. In mask M, positions with value 1 correspond to the patch coverage

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area, while positions with value 0 preserve the original image content. Through this masking mechanism, we can ensure that the adversarial patch only affects specific regions of the target vehicle without affecting other parts of the image, thereby enhancing the concealment and practicality of the attack.

B. Region Sensitivity Analysis

Previous research shows that successful attack regions for fixed-shape patches exhibit spatial clustering patterns. We extend this investigation to examine how shape-optimized patches perform across different models and modalities. This analysis reveals an interesting phenomenon: these patches also demonstrate consistent clustering behavior across various models and modalities, which we term "region sensitivity."

To visually compare the sensitive regions between different modalities on the same UAV vehicle images, we conduct a systematic experiment. We select 200 images from the DroneVehicle dataset [44] and perform comprehensive region sensitivity analysis across multiple advanced YOLO detector versions. Our approach involves applying occlusion-based perturbations to different vehicle regions and measuring the resulting confidence score drops to identify the most vulnerable areas for patch placement. We divide each detected vehicle from the aerial drone perspective into a 3×3 grid, creating nine distinct regions. For each region, we apply complete occlusion (setting pixel values to 0) and measure the confidence score changes across the popular YOLOv5 [45], YOLOv8 [46], YOLOv11 [47], and the newest YOLOv12 [48] detectors in visible and infrared modalities.

To quantify this sensitivity, we define the sensitivity score for region r against model m as

$$S(r,m) = C_{\text{ori}}^{(m)} - C_{\text{adv}}^{(r,m)} \tag{4}$$

where $C_{\text{ori}}^{(m)}$ represents the detection confidence of model m on the original intact image and $C_{\text{adv}}^{(r,m)}$ denotes the detection confidence after occluding region r.

Based on these individual sensitivity scores, we calculate the average importance scores for each modality across all tested models

$$V_{\text{RGB}}(r) = \frac{1}{N_{\text{vis}}} \sum_{m \in M_{\text{vis}}} S(r,m) \tag{5}$$

$$V_{\text{IR}}(r) = \frac{1}{N_{\text{inf}}} \sum_{m \in M_{\text{inf}}} S(r,m) \tag{6}$$

where N_{vis} and N_{inf} represent the number of models in the visible and infrared modalities, respectively, while M_{vis} and M_{inf} denote the corresponding model sets.

The experimental results presented in Fig. 2 reveal clear regional sensitivity patterns. Fig. 2(a) shows that the center region consistently achieves the highest sensitivity scores across all eight model-modality combinations, with values ranging from 0.173 to 0.306, while all peripheral regions remain below 0.1. Fig. 2(b) confirms strong cross-modal consistency: the center region scores 0.240 ($\pm$0.030) in visible modality and 0.240 ($\pm$0.025) in infrared modality. Peripheral regions score much lower, ranging from 0.044 to 0.075, which represents a sensitivity difference of 3.2–5.5 times between central and edge regions.

To further understand this consistency across modalities, we analyze the feature representations of vehicle targets in the two modalities. Although visible and infrared images exhibit different appearance characteristics at lower feature levels, they preserve a similar spatial layout of the same vehicle target. The central roof contains relatively complete target structures, providing shape and texture cues in visible images and thermal radiation cues in infrared images, while peripheral regions contain more background and incomplete object boundaries. During feature extraction at multiple scales, the cues from each modality are concentrated at corresponding spatial locations and provide important information for both category recognition and object localization. Consequently, occluding the central region removes more discriminative information from both modalities and produces a larger confidence decrease than occluding peripheral regions. The results on Faster R-CNN [49] and DETR [50] in Section IV-B further show that this regional pattern remains consistent across different feature extraction and prediction mechanisms.

These findings confirm that the vehicle's central region dominates detection performance universally across YOLO architectures and modalities. Based on these findings, we strategically limit our adversarial patch search space to the vehicle's central region. This approach ensures that adversarial patches target the most vulnerable detection features. By focusing our attack on regions that are 3.2–5.5 times more sensitive, we reduce the search space to just 11% of the vehicle's total area. This strategy maintains attack effectiveness while significantly reducing computational complexity.

C. Grid-Constrained Bezier Patch Generation

Based on the region sensitivity analysis results, adversarial patches achieve optimal effectiveness when placed within the central roof area of vehicles. This finding has driven the development of a grid-constrained approach. The method combines spatial organization with smooth curve generation. A nine-grid modeling system is proposed to achieve precise control over patch generation. The system divides the central roof region into a 3×3 grid structure. Eight outer grid cells serve as anchor and control point locations. The center cell is deliberately excluded, creating an open design space.

The nine-grid modeling provides several technical advantages. The structured spatial constraints prevent patches from extending beyond vehicle roof boundaries, as illustrated by the boundary violation case in Fig. 3(b). The uniform distribution of anchor points across the target region avoids the clustering artifacts shown in Fig. 3(c), where concentrated control points produce unnatural distortions. As a result, the search space is constrained to physically realizable configurations and efficient optimization is achieved.

For smooth patch boundary generation, we employ quadratic Bezier curve interpolation between designated anchor points. Fig. 3(a) demonstrates the effectiveness of this selection. This method provides dual benefits for constraint requirements. First, Bezier curves ensure natural smooth transitions in both visible and infrared imagery and eliminate

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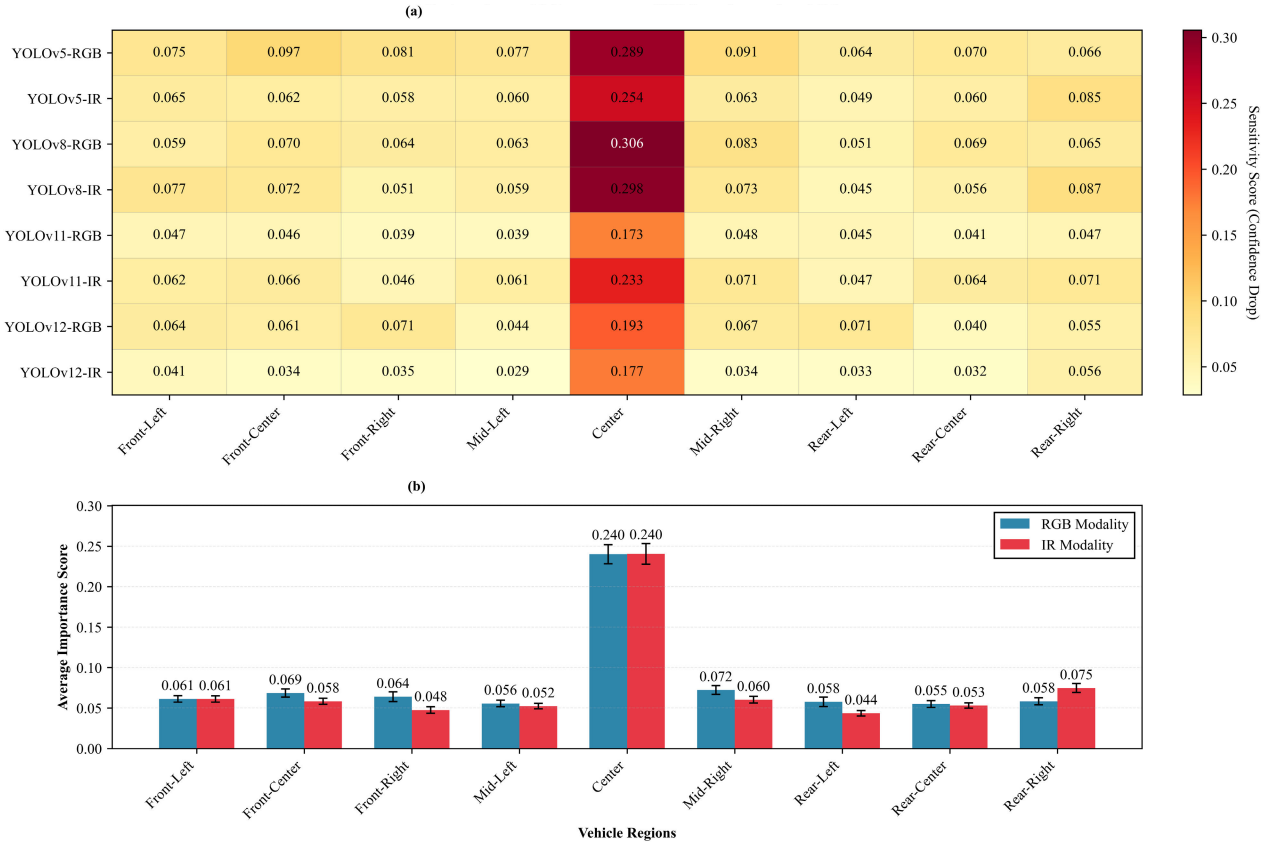


Fig. 2. Region sensitivity analysis for cross-modal UAV vehicle detection. (a) Region-wise sensitivity scores across YOLO versions and modalities. (b) Comparison of region importance between RGB and IR modalities.

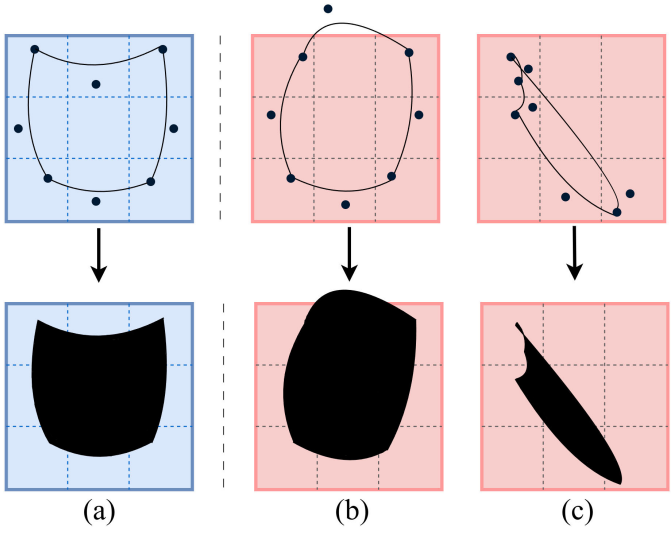


Fig. 3. Grid-constrained patch generation comparison. (a) Optimal configuration: smooth Bezier curve patch under grid constraints. (b) Boundary violation: patch extending beyond target area without spatial constraints. (c) Clustering artifacts: shape distortion caused by control point concentration.

the sharp edges that characterize traditional polygon-based approaches. Second, because the nine-grid system has already constrained points to specific spatial regions, the convex hull property of Bezier curves creates a dual constraint mechanism that guarantees generated curves remain within boundaries defined by their control points.

Our Bezier curve generation follows quadratic interpolation. Given anchor points P_0, P_1, and P_2, the curve segment is defined as

$$B(t) = (1-t)^2 P_0 + 2t(1-t) P_1 + t^2 P_2, t \in [0,1]. \quad (7)$$

The patch generation employs an optimized eight-point configuration where four points serve as curve anchors and four points function as control points. This arrangement creates four connected Bezier curve segments that form a closed contour defining our adversarial patch shape. The complete patch boundary undergoes a fill operation to generate the final binary mask M, where pixels inside the contour receive value 1 (patch region) and pixels outside receive value 0 (background region).

D. Multiobjective Optimization

The cross-modal adversarial attack requires simultaneous optimization of two conflicting objectives: maximizing confidence degradation in both infrared and visible modalities. Traditional single-objective optimization methods cannot effectively handle this multiobjective nature, as improvements in one modality may degrade performance in the other. Therefore, we adopt NSGA-II to solve this multiobjective opti-

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mization problem. The optimization problem can be expressed as follows:

$$\max_s [Z_1(s), Z_2(s)] \tag{8}$$

where

$$Z_1(s) = f_{\text{vis}}(I_{\text{vis}}) - f_{\text{vis}}\left(I_{\text{vis},s}^{\text{adv}}\right) \tag{9}$$

$$Z_2(s) = f_{\text{inf}}(I_{\text{inf}}) - f_{\text{inf}}\left(I_{\text{inf},s}^{\text{adv}}\right) \tag{10}$$

where $Z_1(s)$ and $Z_2(s)$ represent the reduction in detection confidence for the visible and infrared modalities. A larger value indicates a more effective attack. $f_{\text{vis}}(I_{\text{vis},s}^{\text{adv}})$ and $f_{\text{inf}}(I_{\text{inf},s}^{\text{adv}})$ are similar to $f_{\text{vis}}(I_{\text{vis}}^{\text{adv}})$ and $f_{\text{inf}}(I_{\text{inf}}^{\text{adv}})$.

In the NSGA-II approach, both objectives are optimized concurrently without favoring one modality over the other. The following analysis explains why this multiobjective formulation is necessary.

1) Why Multiobjective Formulation Is Necessary: A straightforward alternative is to scalarize the two modality objectives into a single fitness value. For instance, the baseline UAP method [15] adopts a bottleneck (min) operator that takes the lesser of the two modality-specific fitness values as the scalar objective for DE. While intuitive, this scalarization forces the optimizer to focus exclusively on whichever modality is currently weaker and collapses the 2-D objective space into a 1-D search trajectory. As a consequence, the optimizer is unable to independently explore tradeoff regions where moderate improvements in both modalities jointly yield successful cross-modal attacks. In contrast, our multiobjective formulation maintains Z_1 and Z_2 as independent objectives, which allows NSGA-II to discover diverse solutions distributed along the Pareto front. As we demonstrate empirically in Section IV-E, this independent treatment leads to substantially broader coverage of the objective space. NSGA-II solutions spread across a wide range of Z_2 values, whereas DE solutions under bottleneck scalarization remain confined to a narrow low-performance band. Crucially, 86.2% of test samples exhibit nonconvex Pareto fronts, which confirms that the relationship between the two modality objectives is inherently complex and cannot be adequately captured by any single-scalar reduction. The multiobjective formulation also eliminates the need for manual weight tuning between modalities and thus enhances the framework's generalizability across different detector architectures.

2) Individual Representation and Encoding: Each individual represents a candidate adversarial patch solution in the NSGA-II implementation. The encoding utilizes a set of control points for Bezier curve generation. An individual s is formally defined as

$$s = \{(x_1, y_1), (x_2, y_2), \ldots, (x_8, y_8)\}. \tag{11}$$

Each coordinate pair (x_i, y_i) represents a control point within the grid-constrained search space. The eight control points are arranged in clockwise order around the nine-grid structure. These points alternate between anchor points and control points for Bezier curve segments.

Spatial constraints derived from region sensitivity analysis ensure proper positioning. Each control point remains within its designated grid cell. For control point i, the constraints are formulated as

$$x_{\min}^{(i)} \le x_i \le x_{\max}^{(i)}, \quad y_{\min}^{(i)} \le y_i \le y_{\max}^{(i)} \tag{12}$$

where $(x_{\min}^{(i)}, y_{\min}^{(i)})$ and $(x_{\max}^{(i)}, y_{\max}^{(i)})$ define the boundaries of the ith grid cell. This encoding strategy provides multiple advantages. Spatial locality is maintained by constraining points to sensitive regions. Smooth patch boundaries are ensured through Bezier curve interpolation. Efficient genetic operations are enabled while preserving feasibility constraints.

3) Nondominated Sorting and Selection Mechanism: NSGA-II ranks candidate solutions through nondominated sorting based on Pareto dominance relationships. A solution is preferred if it is not worse in any objective and is strictly better in at least one [25]. To maintain diversity along the Pareto front, a crowding distance metric is computed for individuals that share the same dominance rank. The selection mechanism combines these two criteria: individuals with lower rank are always preferred, and among individuals of equal rank, those with larger crowding distance are selected. Detailed mathematical formulations are provided in [25].

4) Adaptive Restart Strategy: Population stagnation presents a major challenge in exploring complex optimization landscapes. To address this issue, an adaptive restart mechanism is employed. Upon detection of convergence stagnation, elite solutions are preserved while the remaining individuals are reinitialized. This mechanism improves the global exploration ability of the algorithm and reduces the likelihood of premature convergence to local optima. The reinitialization of the remaining individuals near the best solution is expressed by the following formula:

$$S_{new}(k) = clip(S_{best}(k) + N(0, \Sigma)) \tag{13}$$

where $S_{best}(k)$ represents the best individual in the population, $N(0, \Sigma)$ denotes a multivariate normal distribution with zero mean and covariance matrix, and $clip(\cdot)$ is a clipping operation that restricts the newly generated individual $S_{new}(k)$ to a certain range, ensuring that it remains within the valid region of the search space.

The complete illustration of this framework is given in Fig. 4.

IV. Experiments

A. Experimental Settings

1) Dataset: Experiments are conducted on the DroneVehicle dataset [44], which comprises five distinct vehicle categories and offers highly synchronized visible and infrared image pairs from aerial perspectives. The dataset contains real-world UAV imagery collected across diverse scenarios including urban roads, residential areas, and parking lots, under illumination conditions that range from daytime to dark night, and at multiple flight altitudes and viewing angles. To construct the attack sample set, we select 100 images from the test set on which the target model achieves high recognition

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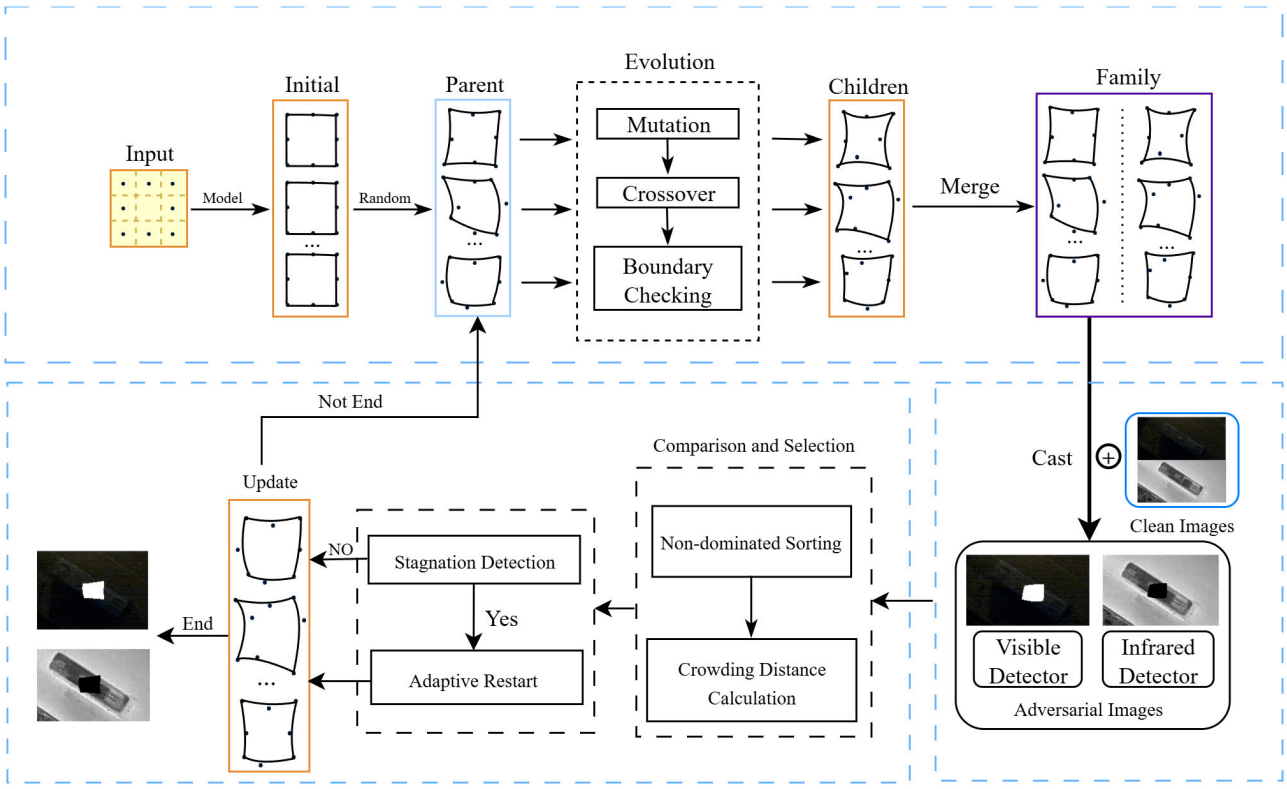


Fig. 4. Overview of BCAP attack framework. The process starts with an initial population of random shapes. Evolution occurs through mutation, crossover, and boundary checking. The parent population is compared to the child population, and individuals with higher fitness are selected, promoting a more balanced and adversarial population. Stagnation detection is used to identify when the population has stopped evolving. When stagnation is detected, an adaptive restart mechanism reinitializes certain individuals, improving diversity and exploration. Finally, the optimal individual is selected to generate the final adversarial samples.

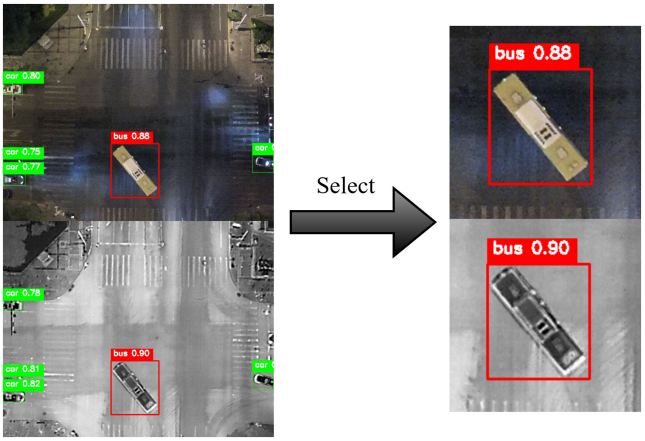


Fig. 5. Target-instance selection on the DroneVehicle dataset. (Left) Detections and confidence scores for the (top) visible and (bottom) infrared images. (Right) Vehicle with the highest average confidence across modalities is selected as the attack target.

accuracy as representative samples. In addition, for category-specific evaluation, we select another 100 images for each vehicle category from the test set. As shown in Fig. 5, when an image contains multiple vehicles, the instance with the highest cross-modal average confidence is selected as the final attack target. This selection strategy has ensured a baseline clean average precision (AP) and mean AP (mAP) of **100%**, thereby providing a reliable reference for subsequent attack performance evaluation.

2) Target Detector: The YOLO series detectors have been widely adopted in UAV remote sensing scenarios due to their superior performance. In this study, we select the established YOLOv5 [45] and YOLOv8 [46] models for evaluation, while expanding our scope to include the newer YOLOv11 [47] and the most advanced YOLOv12 [48]. It is worth noting that, to the best of our knowledge, existing adversarial attack literature has not yet assessed the robustness of YOLOv11 or the state-of-the-art YOLOv12. This study fills this critical gap. For each detector, the official pre-trained weights are used for initialization, and the model is subsequently retrained on the training dataset. These trained models are then employed as target models in the attack process.

3) Metrics: We adopt attack success rate (ASR), AP drop, and mAP drop as metrics to evaluate attack performance. To highlight the effectiveness of simultaneously attacking both modalities, we employ the cross-modal ASR proposed in [15].

This metric measures the proportion of test images that are successfully attacked in both the visible and infrared modalities. The AP drop reflects the change in AP before and after the attack, while the mAP drop quantifies the change in mean AP, thereby assessing the impact of the attack on overall detection performance.

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TABLE I
REGION SENSITIVITY OF THE CENTRAL VEHICLE REGION ACROSS ANCHOR-BASED AND TRANSFORMER-BASED DETECTORS

Model	Modality	Center sensitivity	Center/peripheral mean	Center/peripheral max	Center rank
Faster R-CNN	Infrared	0.5591	6.60	3.31	1
	Visible	0.6863	5.66	3.09	1
DETR	Infrared	0.1712	3.31	2.21	1
	Visible	0.1940	4.00	2.92	1

4) Implementation: As we use the NSGA-II algorithm, we set the number of the initial population to 30 and the epochs of evolution to 130. All hyperparameters are verified on the validation set.

B. Region Sensitivity Validation

To verify that the central dominance is a general property of UAV vehicle targets, we extend the analysis to two detectors with fundamentally different architectures: Faster R-CNN [49], a two-stage anchor-based detector built upon a region proposal network, and DETR [50], an anchor-free Transformer-based detector that formulates detection as a direct set-prediction problem without handcrafted anchors or nonmaximum suppression. We reuse the same 200 images from the DroneVehicle dataset and the identical occlusion-based protocol adopted in Section III-B, ensuring that the detector architecture is the only varying factor across this comparison.

Table I presents the results. For both detectors and both modalities, the central region consistently attains the highest sensitivity score among all nine regions, yielding a center rank of 1 in every configuration. The central sensitivity exceeds the average of the eight peripheral regions by factors of 3.31–6.60, comparable to the 3.2–5.5 times difference observed on the YOLO detectors in Section III-B, and it remains 2.21–3.31 times higher than even the single most sensitive peripheral region. Although the absolute sensitivity magnitudes vary across architectures, the relative dominance of the central region is consistently preserved. This dominance holds for the anchor-free DETR as well, whose set-prediction mechanism introduces no anchor priors toward the geometric center, indicating that the central vulnerability arises from the concentration of discriminative semantic cues, such as the roof structure and window contours, rather than from the anchor assignment of any specific detector.

Based on the above observations, we further evaluate the practical value of the regional sensitivity analysis described in Section III-B; we compare two patch placement strategies. The first strategy covers the entire rooftop of the target vehicle with the patch, while the second restricts the patch to the central rooftop region. Both strategies are tested under identical optimization settings to ensure comparability. Visual examples of both placement strategies are provided in Fig. 6, further illustrating the contrast in spatial coverage.

Table II presents the quantitative results demonstrating the effectiveness of central roof placement over full roof coverage. When the patch is restricted to the central rooftop region, the cross-modal ASR increases marginally from 59% to 60%,

TABLE II
COMPARISON OF ATTACK PERFORMANCE AND EFFICIENCY BETWEEN FULL ROOF AND CENTRAL ROOF PATCH PLACEMENT

Region	*ASR*(%)			Avg query
	Cross-Modal	Visible	Infrared	
Full	59%	74%	78%	2703.6
Central	**60%**	**77%**	71%	**1729.2**

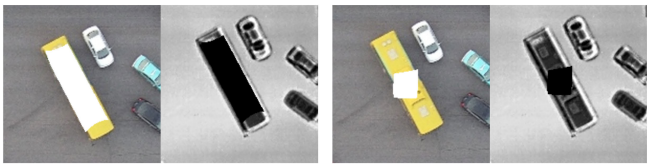


Fig. 6. Adversarial patch examples for two placement strategies in visible and infrared modalities. (a) Full roof placement. (b) Central roof placement.

while the visible ASR rises from 74% to 77%. Although the infrared ASR decreases from 78% to 71%, the computational efficiency improvements are substantial, with the average number of queries reduced from 2703.6 to 1729.2, which represents a 36.02% computational cost reduction. Notably, to achieve comparable ASRs when employing full roof placement, the optimization requires increasing epochs from 130 to 280 and expanding the population number by 10, which further demonstrates the computational advantage of central region targeting. These results validate that strategic patch placement in high-sensitivity regions maintains attack effectiveness while delivering substantial computational efficiency gains.

C. Cross-Modal Attack Performance Evaluation

To evaluate BCAP's effectiveness against state-of-the-art detection systems, comprehensive experiments are conducted across four advanced YOLO architectures. We specifically select UAP [15] as the baseline comparison method, as it represents the sole existing method capable of performing black-box adversarial attacks against visible-infrared cross-modal object detection. While other cross-modal attack approaches exist, they assume white-box access, precluding a fair comparison within our black-box threat model. The original UAP framework employs a dual-patch configuration for ground-level targets. However, UAV vehicle detection presents unique constraints due to small target sizes and limited roof surface area when viewed from aerial perspectives. These geometric limitations necessitate modification of UAP to a single-patch configuration to ensure feasible patch placement within the target boundaries. HCB [17] represents an

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TABLE III
ATTACK PERFORMANCE COMPARISON ON DIFFERENT OBJECT DETECTION SYSTEMS

Model	Method	*ASR*(%)			*mAP* drop	
		Cross-Modal	Visible	Infrared	Visible	Infrared
YOLOv5	UAP	37%	46%	52%	29.99%	42.80%
	BCAP	**54%**	**71%**	**70%**	**48.38%**	**55.82%**
YOLOv8	UAP	42%	53%	45%	38.98%	34.55%
	BCAP	**61%**	**78%**	**73%**	**54.72%**	**53.03%**
YOLOv11	UAP	38%	50%	43%	33.23%	29.07%
	BCAP	**53%**	**73%**	**68%**	**48.73%**	**49.07%**
YOLOv12	UAP	37%	50%	51%	33.93%	36.63%
	BCAP	**50%**	**70%**	**67%**	**48.72%**	**46.95%**

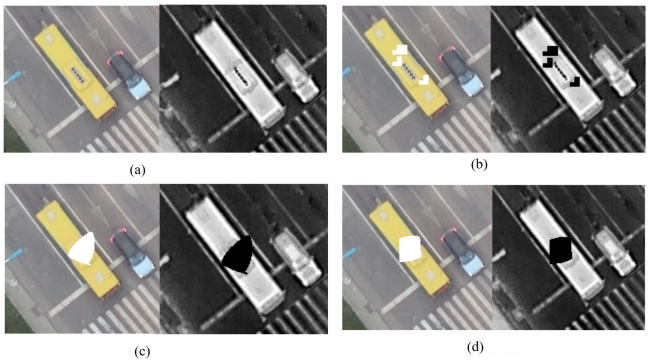


Fig. 7. Qualitative comparison of patch generation methods on visible-infrared image pairs. (a) Clean images without adversarial perturbations. (b) HCB patches optimized for infrared modality only. (c) UAP baseline patches with irregular boundaries. (d) BCAP patches generated by grid-constrained Bezier curves.

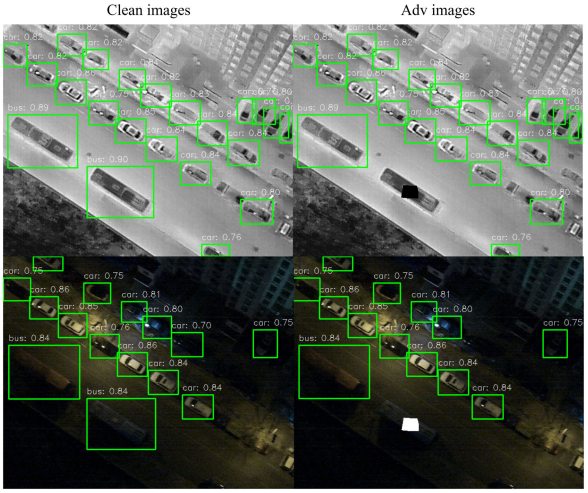


Fig. 8. Cross-modal attack examples generated by BCAP.

infrared-specific attack method and is excluded from cross-modal comparisons due to its single-modality limitation.

Fig. 7 presents visual comparisons of different patch generation approaches across visible and infrared modalities. BCAP produces patches with smoother boundaries compared to UAP's irregular shapes, a result of the grid-constrained Bezier curve generation method. Beyond visual differences, the primary advantage of this parameterization lies in providing a continuous, low-dimensional search space that facilitates more efficient optimization. The quantitative contribution of this search space design is analyzed in Section IV-E.

Attack effectiveness is demonstrated in Fig. 8, where clean images achieve consistent high-confidence vehicle detection, whereas BCAP-generated adversarial samples successfully evade detection. The target vehicle in the clean infrared image exhibits a confidence score of 0.84, which drops below the detection threshold after patch application, resulting in complete evasion.

Table III indicates that our method outperforms UAP across all four detector models, achieving cross-modal ASRs of 50%–61% compared to UAP's 37%–42% range. Our method is equally effective for single-modality scenarios, achieving 70%–78% ASRs in visible modality compared to UAP's 46%–53%. Similarly, infrared modality performance reaches 67%–73% versus UAP's 43%–52%.

Detection degradation metrics further validate BCAP's attack potency, with mAP drops substantially exceeding UAP's performance across both modalities. Visible modality achieves 48%–55% mAP reduction compared to UAP's 30%–39%, while infrared modality reaches 47%–56% versus UAP's 29%–43%. Even against the state-of-the-art YOLOv12, which incorporates the latest attention mechanisms and architectural improvements, BCAP maintains effective attack performance. The comprehensive evaluation demonstrates that BCAP generalizes well, as it delivers consistent performance gains regardless of architectural complexity.

D. Category-Specific Attack Analysis

BCAP demonstrates superior performance across all vehicle categories compared to UAP. Table IV presents comprehensive attack results on YOLOv8 across five vehicle types. Cross-modal ASR shows consistent advantages for BCAP, with performance varying significantly by vehicle category. Truck achieves the highest success rate at 89% versus UAP's 67%, while freight car improves from 53% to 72%. Car shows the largest relative gain, rising from 30% to 61%. Bus improves from 38% to 58%, and van increases from 58% to 61%.

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TABLE IV
CATEGORY-SPECIFIC ATTACK PERFORMANCE ON YOLOV8

Class	Method	*ASR*(%)			*mAP* drop	
		Cross-Modal	Visible	Infrared	Visible	Infrared
Car	UAP	30%	50%	33%	45.45%	18.18%
	BCAP	**61%**	**80%**	**63%**	**72.73%**	**45.45%**
Truck	UAP	67%	73%	74%	49.66%	63.64%
	BCAP	**89%**	**92%**	**95%**	**59.73%**	**72.73%**
Bus	UAP	38%	56%	42%	54.55%	36.36%
	BCAP	**58%**	**82%**	**69%**	**72.73%**	**63.64%**
Van	UAP	58%	71%	62%	50%	54.55%
	BCAP	**61%**	**84%**	**67%**	**60%**	54.55%
Freight car	UAP	53%	66%	70%	50%	63.88%
	BCAP	**72%**	**75%**	**92%**	**60%**	**81.82%**

TABLE V
COMPARISON OF PATCH GENERATION METHODS UNDER THE SAME NSGA-II OPTIMIZER

Method	*ASR*(%)			*mAP* drop		Avg query
	Cross-Modal	Visible	Infrared	Visible	Infrared	
UAP	49%	72%	60%	43.64%	45.54%	4358
BCAP	**60%**	**77%**	**71%**	**56.69%**	**52.73%**	**1804.8**

Visible modality results follow similar patterns. BCAP achieves 75%–92% ASR across categories, while UAP ranges from 50% to 73%. Truck appears most susceptible in visible modality, reaching 92%. Car demonstrates substantial improvement from 50% to 80%. Infrared results exhibit category-dependent vulnerability patterns. Bus attains 69% and truck reaches 95% under BCAP, both substantially exceeding UAP performance. mAP degradation confirms attack effectiveness, with infrared drops of 45%–82% observed under BCAP versus 18%–64% for UAP. Freight car shows the largest degradation at 81.82%, while car rises from 18.18%–45.45%.

These category-dependent variations stem from distinct vehicle characteristics and thermal properties. Large vehicles provide wider, flatter central roofs where patch-to-box area ratios are higher and projections remain stable. Simpler textures and stronger, more uniform thermal emission further increase sensitivity to structured perturbations in both modalities. In contrast, cars present smaller targets with varying roof shapes and weaker thermal contrast, which reduces both visible and infrared attack effectiveness. These physical constraints explain the performance differences across vehicle categories.

These results confirm that BCAP adapts effectively to diverse vehicle characteristics and maintains consistent improvements across all tested categories in both visible and infrared modalities.

E. Analysis of Patch Generation and Optimization Strategy

This section validates two independent design decisions of BCAP: the patch generation method, where grid-constrained Bezier curves are compared with UAP parameterization, and the optimization strategy, where scalar formulations are compared with multiobjective formulations. By decoupling these two factors, we can precisely quantify the individual contribution of each component. All experiments in this section are conducted on YOLOv8 to ensure controlled comparison.

1) Patch Generation Method Comparison: To evaluate the patch generation method independently of the optimizer choice, we compare two parameterization methods under the same NSGA-II optimizer with identical hyperparameters: UAP's original patch representation and our grid-constrained Bezier curve generation. Both methods operate within the central vehicle roof region to ensure spatial fairness.

Table V presents the results. Under the same optimizer, our Bezier generation achieves 60% cross-modal ASR compared to UAP's 49%, which represents an 11 percentage point improvement. Visible ASR increases from 72% to 77%, and infrared ASR increases from 60% to 71%. The average query count also drops from 4358 to 1804.8. These results indicate that the patch generation method yields consistent performance improvements when the optimizer is held constant.

A cross-comparison with Table VI provides additional evidence. UAP with NSGA-II achieves only 49% cross-modal ASR, and our Bezier method with the weaker single-objective DE optimizer achieves 52%. Neither a superior optimizer alone nor an improved parameterization alone exceeds 52%. However, when the two are combined, cross-modal ASR reaches 60%, which demonstrates that the patch generation method and the multiobjective optimizer are complementary and that their combination produces gains beyond what either component achieves independently.

2) Optimization Strategy Comparison: Having established the effectiveness of our patch generation method, we further examine the choice of optimization strategy under the BCAP

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TABLE VI
COMPARISON OF OPTIMIZATION STRATEGIES UNDER THE BCAP FRAMEWORK

Optimizer	*ASR*(%)			*mAP* drop		Avg query
	Cross-Modal	Visible	Infrared	Visible	Infrared	
DE	52%	57%	55%	38.98%	45.45%	3342.8
DEMO	57%	73%	67%	50.79%	49.09%	2719.2
CMF-NSGA-II	**62%**	77%	71%	54.72%	50.91%	2765.1
NSGA-II	60%	77%	71%	**56.69%**	**52.73%**	**1804.8**

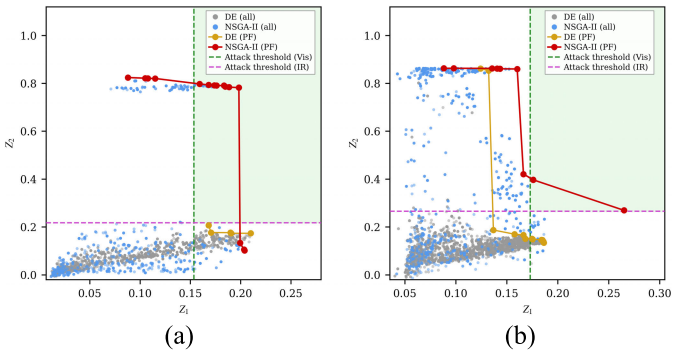


Fig. 9. Comparison of search trajectories between NSGA-II and DE in the objective space for two representative samples. (a) L-shaped nonconvex Pareto front. (b) Broader search coverage of NSGA-II over DE. The green shaded region indicates the cross-modal attack success area.

framework. Table VI compares four optimization configurations with progressive complexity. In addition to the standard DE and NSGA-II, we include DEMO [51], a multiobjective extension of DE that replaces scalar fitness with Pareto selection, while it retains the same DE mutation and crossover operators.

The first comparison between DE and DEMO isolates the effect of objective formulation. The only difference is the objective treatment: DE aggregates both modality objectives into a single scalar through a bottleneck operator that takes the minimum of the two fitness values, whereas DEMO optimizes them as independent objectives. As shown in Table VI, cross-modal ASR improves from 52% to 57%, visible ASR rises substantially from 57% to 73%, and infrared ASR increases from 55% to 67%. These improvements indicate that the two modality objectives are inherently conflicting and benefit from independent optimization rather than scalar aggregation.

The second comparison between DEMO and NSGA-II evaluates different multiobjective solvers under the same formulation. NSGA-II achieves 60% cross-modal ASR compared to DEMO's 57%, with visible ASR at 77% and infrared ASR at 71%. The average query count decreases from 2719.2 to 1804.8, which represents a 33.6% reduction. These performance gains are consistent with the nondominated sorting and crowding distance mechanisms in NSGA-II, which allow broader coverage of the objective space.

Fig. 9 compares the search trajectories of NSGA-II and DE in the objective space for two representative samples. The figure contrasts the strongest multiobjective solver against the scalar baseline to illustrate why independent objective treatment leads to broader search coverage. The green shaded region indicates the cross-modal attack success area where both modality confidence scores drop below the detection threshold. In Fig. 9(a), the NSGA-II Pareto front shown as the red curve exhibits a clear L-shaped nonconvex pattern. Z_2 remains high between 0.78 and 0.82 before it drops sharply to approximately 0.10. Multiple NSGA-II solutions fall within the success region, whereas all DE solutions shown as gray dots fail to enter it despite nearly double the query budget, approximately 1880 compared to 1050. The NSGA-II search population shown as blue dots spreads broadly across $Z_2 > 0.3$, while DE solutions remain confined below $Z_2 < 0.2$. The bottleneck scalarization restricts the optimizer's exploration of the infrared objective dimension. Fig. 9(b) confirms these patterns on a different sample, where DE consumes 8560 queries yet produces no successful solutions, while NSGA-II succeeds with approximately 2700 queries. Across all 100 test images, 86.2% exhibit nonconvex Pareto fronts. This prevalence indicates that the relationship between visible and infrared modality objectives is inherently complex and cannot be adequately captured by any scalar reduction.

The third comparison examines CMF-NSGA-II, which extends standard NSGA-II with a cross-modal focus mechanism that dynamically adjusts selection pressure to prioritize individuals with simultaneous performance in both modalities. Unlike standard NSGA-II, CMF-NSGA-II does not treat the two objectives with equal weight at all stages but instead allocates more selection pressure toward cross-modal coordination. CMF-NSGA-II raises cross-modal ASR to 62%, but the query count increases to 2765.1. The mAP drop metrics reveal a notable tradeoff. CMF-NSGA-II achieves 54.72% visible mAP drop and 50.91% infrared mAP drop, both lower than NSGA-II's 56.69% and 52.73%, respectively. This tradeoff indicates that more aggressive cross-modal focus improves cross-modal ASR but sacrifices overall detection degradation and incurs higher computational cost. Standard NSGA-II achieves a favorable balance between attack effectiveness and computational efficiency.

3) Restart Strategy Evaluation: Table VII compares performance with and without the adaptive restart mechanism. The restart mechanism reduces average queries from 1911.3 to 1729.2, which represents a 9.53% efficiency improvement. Attack performance remains consistent across modalities, with cross-modal ASR stable at 60% and visible modality at 77%.

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TABLE VII
ADAPTIVE RESTART STRATEGY IMPACT ON ATTACK PERFORMANCE AND COMPUTATIONAL EFFICIENCY

Configuration	ASR(%)			Avg query
	Cross-Modal	Visible	Infrared	
Without Restart	60%	77%	70%	1911.3
With Restart	60%	77%	**71%**	**1729.2**

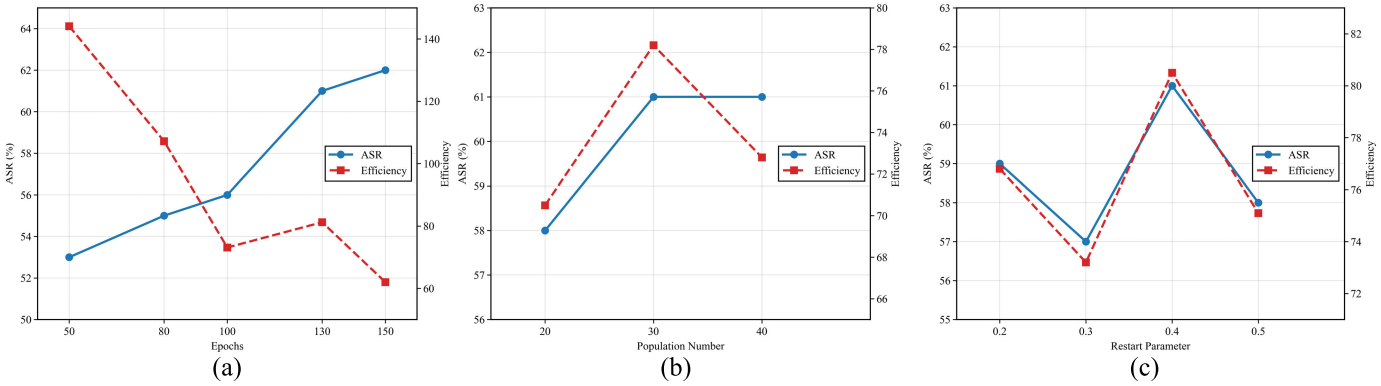


Fig. 10. Parameters tuning in our method. (a) Effect of the number of epochs. (b) Effect of the population number. (c) Effect of the restart parameter.

Infrared modality improves marginally from 70% to 71%. Computational time is reduced by 19.23%. These results indicate that the adaptive restart strategy enhances computational efficiency without any compromise in attack quality.

The analyses above lead to two independent conclusions.

First, the grid-constrained Bezier patch generation contributes an 11 percentage point cross-modal ASR improvement when the optimizer is held constant, as demonstrated in Table V. Second, multiobjective formulation is necessary for cross-modal attacks, as supported by the quantitative comparison between scalar and multiobjective configurations in Table VI and the prevalence of nonconvex Pareto fronts across 86.2% of test samples in Fig. 9. Within the multiobjective paradigm, NSGA-II achieves a favorable balance between attack effectiveness and computational efficiency. These 2-D jointly underpin the design of the BCAP framework.

F. Parameter and Ablation Studies

In this section, we tune hyperparameters against the YOLOv8 model. To comprehensively evaluate the performance of different parameter configurations, we propose a comprehensive efficiency metric E that balances the tradeoff between ASR and computational resource consumption. The comprehensive efficiency E is defined as the ratio of ASR to weighted normalized resource cost. The formulation can be expressed as follows:

$$E = \frac{ASR}{\alpha^* T_{norm} + \beta^* Q_{norm}} \tag{14}$$

where T_{norm} and Q_{norm} represent the normalized execution time and query count, respectively. The weight coefficients α and β are set to 0.7 and 0.3, which reflects the relative importance of execution time in practical applications. It should be noted that E is not intended as an absolute performance measure, but rather as an auxiliary criterion to assist the selection of hyperparameter configurations in this section. Therefore, the specific weight ratio is a configurable design choice based on practical deployment requirements, and it can be adjusted according to different scenarios.

1) Epoch: The epochs represent the maximum number of evolutionary iterations for a single image. Thus, to explore the sensitivity of our method, we set epochs as 50, 80, 100, 130, and 150. Fig. 10(a) reveals distinct patterns in both ASR and efficiency as epoch count increases from 50 to 150. Cross-modal ASR rises steadily from 58% at 50 epochs to 62% at 130 epochs. Beyond this point, performance plateaus with minimal improvement at 150 epochs. Computational efficiency drops significantly after 130 epochs, while performance gains become marginal. Based on these observations, 130 epochs have been selected as the optimal setting. This configuration achieves a favorable tradeoff between attack effectiveness and efficiency.

2) Population Number: The number of populations affects the population diversity. A small number of populations means that the populations do not differ much and are easy to fall into the local optimal solution. However, larger populations do not necessarily yield better results. Excessive population sizes increase per-generation computational cost without proportional performance gains. The algorithm requires more iterations to reach stable solutions when population size increases beyond optimal thresholds. As demonstrated in Fig. 10(b), when the population size increases from 20 to 30, the cross-modal ASR improves substantially from 58% to 62%, representing a significant four percentage point gain. However, further increasing the population to 40 yields only marginal improvement (1% increase) while substantially increasing computational cost. Additionally, we observe that efficiency remains relatively stable between 30 and 40 individ-

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uals, indicating diminishing returns beyond the optimal size. Therefore, to maintain an effective balance between solution quality and computational feasibility, we set the population size to 30.

3) Restart Parameter: The restart parameter determines the proportion of elite individuals preserved during adaptive restart mechanisms. This balance proves essential for maintaining exploration-exploitation tradeoffs. We evaluate parameter values from 0.2 to 0.5. Fig. 10(c) presents the quantitative results. Optimal performance occurs at 0.4, which achieves 61% ASR. At 0.2, excessive reinitialization causes loss of valuable genetic information and reduces convergence stability. At 0.5, too many individuals are preserved, which limits exploration of new solution regions. The value 0.4 balances exploration and exploitation and is selected as the final setting.

G. Robustness Under Simulated Environmental Variations

To evaluate the robustness of BCAP patches under potential environmental variations, we apply three types of simulated perturbations to the generated adversarial samples: brightness variation ($\pm$10% and $\pm$20%), additive Gaussian noise (σ = 5, 10), and partial patch occlusion (10%, 20%, and 30% of the patch area). Brightness and noise are applied to the entire image because these factors affect the complete scene in practical UAV operations. Occlusion covers contiguous rectangular blocks within the patch region with neutral gray pixels, which simulates physical obstructions such as shadows or debris on the vehicle roof.

Because environmental perturbations may also degrade detector sensitivity on unattacked images, we adopt a strict sample selection protocol. Starting from the 60 images where cross-modal attack originally succeeded, we apply all six brightness and noise conditions to their clean counterparts and retain only those images where the detector identifies the target in both modalities under every condition. This process yields 43 evaluable samples, all of which achieve 100% ASR across three metrics under no perturbation.

Table VIII presents the results. Under moderate perturbations ($-$10% brightness and σ = 5), the cross-modal ASR remains at 62.8% and 67.4%, respectively, and the infrared ASR stays above 88%, which indicates that the adversarial patterns retain sufficient strength despite global image-level changes. A notable observation is that the infrared modality consistently exhibits higher robustness than the visible modality under brightness variations. For example, at $-$20% brightness the infrared ASR is 81.4% while the visible ASR drops to 58.1%. This asymmetry arises because infrared imaging relies on thermal radiation rather than reflected light, so brightness scaling has a limited effect on the infrared adversarial pattern. Under stronger perturbations (+20% brightness and σ = 10), the visible ASR decreases to 39.5% and 65.1%, respectively, while the cross-modal ASR drops to 34.9% and 46.5%. The cross-modal metric declines more steeply because it requires simultaneous evasion in both modalities, so per-modality degradation compounds under the joint success criterion.

For partial occlusion, ASR decreases progressively as the occluded area increases. At 10% occlusion the cross-modal ASR is 37.8%, and at 30% it decreases to 24.4%. Since BCAP patches occupy only approximately 1.1% of the total image area, even 10% block occlusion removes a proportionally significant portion of the adversarial pattern. Nevertheless, the infrared ASR remains above 64% at 30% occlusion, which confirms that the remaining patch area still provides sufficient perturbation for individual modalities. These results demonstrate that BCAP patches maintain practical robustness under simulated environmental variations.

TABLE VIII
ASR (%) UNDER ENVIRONMENTAL PERTURBATIONS (N = 43). NOISE DENOTES ADDITIVE GAUSSIAN NOISE

Perturbation	Level	*ASR*(%)		
		Cross-Modal	Visible	Infrared
Brightness	-20%	41.9%	58.1%	81.4%
Brightness	-10%	62.8%	74.4%	88.4%
Brightness	+10%	40%	44.4%	84.4%
Brightness	+20%	34.9%	39.5%	76.7%
Noise	$\sigma = 5$	67.4%	76.7%	90.7%
Noise	σ = 10	46.5%	65.1%	79.1%
Occlusion	10%	37.8%	57.8%	68.9%
Occlusion	20%	28.9%	53.3%	71.1%
Occlusion	30%	24.4%	46.7%	64.4%

V. CONCLUSION

In this article, we proposed BCAP, a novel black-box adversarial patch attack framework designed for cross-modal UAV vehicle detectors. By integrating NSGA-II, the framework resolves the optimization conflict between visible and infrared modalities in black-box scenarios. Our region sensitivity analysis identified the central roof as the most vulnerable area across YOLO architectures. Accordingly, we developed a grid-constrained Bezier curve generation method to restrict patches to these sensitive regions while maintaining smooth boundaries. Furthermore, an adaptive restart strategy was integrated to enhance global search capabilities and computational efficiency. Robustness evaluation under simulated environmental variations demonstrates that BCAP patches maintain practical attack effectiveness under brightness changes, additive noise, and partial occlusion, with infrared modality exhibiting consistently higher resilience due to its reliance on thermal radiation. Experimental results on the DroneVehicle dataset demonstrate the superior effectiveness and robustness of BCAP compared to baseline approaches.

However, this study is limited by the absence of physical-world validation, although the use of the real-world DroneVehicle dataset supports the practical relevance of the findings. In addition, the current dataset does not include images captured under extreme weather conditions such as heavy rain or fog, which may affect the generalizability of the results. Moreover, the present evaluation does not account for real-flight UAV dynamics, such as motion blur, rapid altitude variations, and abrupt changes in viewing angle, which may further reduce the cross-modal ASR in practical aerial missions. Future work will focus on physical-world validation, developing adaptive

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patch placement strategies for different viewing perspectives, and extending the evaluation to datasets that cover a broader range of environmental conditions.

REFERENCES

[1] A. Roca et al., "Efficient endangered deer species monitoring with UAV aerial imagery and deep learning," in *Proc. IEEE Biennial Congr. Argentina (ARGENCON)*, Sep. 2024, pp. 1–8.

[2] B. Kellenberger, D. Marcos, S. Lobry, and D. Tuia, "Half a percent of labels is enough: Efficient animal detection in UAV imagery using deep CNNs and active learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 12, pp. 9524–9533, Dec. 2019.

[3] H. Zhu et al., "Intelligent agriculture: Deep learning in UAV-based remote sensing imagery for crop diseases and pests detection," *Frontiers Plant Sci.*, vol. 15, Oct. 2024, Art. no. 1435016.

[4] C. Lyu, S. Lin, A. Lynch, Y. Zou, and M. Liarokapis, "UAV-based deep learning applications for automated inspection of civil infrastructure," *Autom. Construction*, vol. 177, Sep. 2025, Art. no. 106285.

[5] C. Szegedy et al., "Intriguing properties of neural networks," 2013, *arXiv:1312.6199*.

[6] I. J. Goodfellow, J. Shlens, and C. Szegedy, "Explaining and harnessing adversarial examples," 2014, *arXiv:1412.6572*.

[7] K. Eykholt et al., "Robust physical-world attacks on deep learning visual classification," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 1625–1634.

[8] X. Zhu, X. Li, J. Li, Z. Wang, and X. Hu, "Fooling thermal infrared pedestrian detectors in real world using small bulbs," in *Proc. AAAI Conf. Artif. Intell.*, vol. 35, no. 4, 2021, pp. 3616–3624.

[9] P. Zhang, J. Zhao, D. Wang, H. Lu, and X. Ruan, "Visible-thermal UAV tracking: A large-scale benchmark and new baseline," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 8876–8885.

[10] Y. Zhang et al., "Drone-based RGBT tiny person detection," *ISPRS J. Photogramm. Remote Sens.*, vol. 204, pp. 61–76, Oct. 2023.

[11] Y. Zhang et al., "Adversarial patch attack on multi-scale object detection for UAV remote sensing images," *Remote Sens.*, vol. 14, no. 21, p. 5298, Oct. 2022.

[12] G. Tang, W. Yao, C. Li, T. Jiang, and S. Yang, "Black-box adversarial patch attacks using differential evolution against aerial imagery object detectors," *Eng. Appl. Artif. Intell.*, vol. 137, Nov. 2024, Art. no. 109141.

[13] A. Du et al., "Physical adversarial attacks on an aerial imagery object detector," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2022, pp. 1796–1806.

[14] Y. Su, H. Li, and M. Gong, "Generative adversarial patches for physical attacks on cross-modal pedestrian re-identification," 2024, *arXiv:2410.20097*.

[15] X. Wei, Y. Huang, Y. Sun, and J. Yu, "Unified adversarial patch for visible-infrared cross-modal attacks in the physical world," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 4, pp. 2348–2363, Apr. 2024.

[16] C. Hu, W. Shi, W. Yao, T. Jiang, L. Tian, and W. Li, "Two-stage optimized unified adversarial patch for attacking visible-infrared cross-modal detectors in the physical world," *Appl. Soft Comput.*, vol. 171, Mar. 2025, Art. no. 112818.

[17] H. Wei et al., "Hotcold block: Fooling thermal infrared detectors with a novel wearable design," in *Proc. AAAI Conf. Artif. Intell. (AAAI)*, 2023, pp. 15233–15241.

[18] X. Wei, Y. Guo, and J. Yu, "Adversarial sticker: A stealthy attack method in the physical world," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 3, pp. 2711–2725, Mar. 2023.

[19] C. Hu et al., "Adversarial infrared blocks: A multi-view black-box attack to thermal infrared detectors in physical world," *Neural Netw.*, vol. 175, Jul. 2024, Art. no. 106310.

[20] J. Lian, S. Mei, S. Zhang, and M. Ma, "Benchmarking adversarial patch against aerial detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5634616.

[21] W. Ma et al., "Infrared and visible image fusion technology and application: A review," *Sensors*, vol. 23, no. 2, p. 599, Jan. 2023.

[22] J. Chen, M. I. Jordan, and M. J. Wainwright, "HopSkipJumpAttack: A query-efficient decision-based attack," in *Proc. IEEE Symp. Secur. Privacy (SP)*, May 2020, pp. 1277–1294.

[23] A. Ilyas, L. Engstrom, A. Athalye, and J. Lin, "Black-box adversarial attacks with limited queries and information," in *Proc. 35th Int. Conf. Mach. Learn.*, 2018, pp. 2137–2146.

[24] X. Wei, S. Ruan, Y. Dong, H. Su, and X. Cao, "Distributionally location-aware transferable adversarial patches for facial images," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 47, no. 4, pp. 2849–2864, Apr. 2025.

[25] K. Deb, A. Pratap, S. Agarwal, and T. Meyarivan, "A fast and elitist multiobjective genetic algorithm: NSGA-II," *IEEE Trans. Evol. Comput.*, vol. 6, no. 2, pp. 182–197, Feb. 2002.

[26] A. Madry, A. Makelov, L. Schmidt, D. Tsipras, and A. Vladu, "Towards deep learning models resistant to adversarial attacks," 2017, *arXiv:1706.06083*.

[27] N. Carlini and D. Wagner, "Towards evaluating the robustness of neural networks," in *Proc. IEEE Symp. Secur. Privacy (SP)*, May 2017, pp. 39–57.

[28] T. B. Brown, D. Mané, A. Roy, M. Abadi, and J. Gilmer, "Adversarial patch," 2017, *arXiv:1712.09665*.

[29] K. Xu et al., "Adversarial t-shirt! evading person detectors in a physical world," in *Proc. Eur. Conf. Comput. Vis. Cham*, Switzerland: Springer, 2020, pp. 665–681.

[30] M. Sharif, S. Bhagavatula, L. Bauer, and M. K. Reiter, "Accessorize to a crime: Real and stealthy attacks on state-of-the-art face recognition," in *Proc. ACM SIGSAC Conf. Comput. Commun. Secur.*, 2016, pp. 1528–1540.

[31] H. Jiang, H. Han, M. Zhou, and L. Huang, "Transferable adversarial camouflage attacks with multi-layer topology tuning for intelligent vehicles," *IEEE Trans. Veh. Technol.*, vol. 75, no. 6, pp. 9601–9615, Jun. 2026.

[32] R. Yao, A. Zhang, Y. Zhou, J. Zhao, B. Liu, and A. El Saddik, "Adversarial geometric attacks for 3D point cloud object tracking," *IEEE Trans. Multimedia*, vol. 27, pp. 3144–3157, 2025.

[33] X. Zhu, Z. Hu, S. Huang, J. Li, and X. Hu, "Infrared invisible clothing: Hiding from infrared detectors at multiple angles in real world," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 13317–13326.

[34] C. Hu et al., "Adversarial infrared curves: An attack on infrared pedestrian detectors in the physical world," *Neural Netw.*, vol. 178, Oct. 2024, Art. no. 106459.

[35] J.-J. Huang et al., "DeMPAA: Deployable multi-mini-patch adversarial attack for remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5623613.

[36] J. Chen, Y. Zhang, C. Liu, K. Chen, Z. Zou, and Z. Shi, "Digital-to-physical visual consistency optimization for adversarial patch generation in remote sensing scenes," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5623017.

[37] R. Jia, R. He, P. Yue, C. Sun, and W. Sun, "CCA: A novel camouflage coating attack on object detectors in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5631516.

[38] T. Hu et al., "Boosting HDR image reconstruction via semantic knowledge transfer," *IEEE Trans. Image Process.*, vol. 35, pp. 1910–1922, 2026.

[39] Z. Wei, J. Chen, Z. Wu, and Y.-G. Jiang, "Adaptive cross-modal transferable adversarial attacks from images to videos," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 5, pp. 3772–3783, May 2024.

[40] T. Kim, H. J. Lee, and Y. M. Ro, "Map: Multispectral adversarial patch to attack person detection," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, May 2022, pp. 4853–4857.

[41] T. Kim, Y. Yu, and Y. M. Ro, "Multispectral invisible coating: Laminated visible-thermal physical attack against multispectral object detectors using transparent low-E films," in *Proc. AAAI Conf. Artif. Intell. (AAAI)*, 2023, vol. 37, no. 1, pp. 1151–1159.

[42] Y. Liu, T. Song, T. He, Z. Lan, and Q. Liang, "Cross-modal adversarial patch attack based on class activation maps," in *Proc. Int. Conf. Image, Vis. Intell. Syst. (ICVIS)*, 2026, pp. 618–626.

[43] J. Long et al., "CDUPatch: Color-driven universal adversarial patch attack for dual-modal visible-infrared detectors," in *Proc. 33rd ACM Int. Conf. Multimedia*, Oct. 2025, pp. 1462–1470.

[44] Y. Sun, B. Cao, P. Zhu, and Q. Hu, "Drone-based RGB-infrared cross-modality vehicle detection via uncertainty-aware learning," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 32, no. 10, pp. 6700–6713, Oct. 2022.

[45] G. Jocher, "Ultralytics/YOLOv5: V7. 0-YOLOv5 SOTA realtime instance segmentation," Zenodo, CERN, Eur. Org. Nucl. Res., Geneva, Switzerland, 2022, doi: 10.5281/zenodo.7347926.

[46] G. Jocher, A. Chaurasia, and J. Qiu. (2023). *Ultralytics YOLOv8*. [Online]. Available: https://github.com/ultralytics/ultralytics

[47] G. Jocher and J. Qiu. (2024). *Ultralytics YOLO11*. [Online]. Available: https://github.com/ultralytics/ultralytics

[48] Y. Tian, Q. Ye, and D. Doermann, "YOLOv12: Attention-centric real-time object detectors," 2025, *arXiv:2502.12524*.

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[49] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 6, pp. 1137–1149, Jun. 2017.
[50] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, "End-to-end object detection with transformers," in *Proc. Eur. Conf. Comput. Vis. Cham*, Switzerland: Springer, 2020, pp. 213–229.
[51] T. Robič and B. Filipič, "DEMO: Differential evolution for multiobjective optimization," in *Proc. 3rd Int. Conf. Evol. Multi-Criterion Optim.*, 2005, pp. 520–533.



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